ENEE 132 — Basic Electronics I (Semiconductor Devices)

Lecture 6 - The Zener Diode & Applications

14 July 2026

Today's Agenda — Lecture Notes pp. 75–89

- **Part 1 — The Zener Diode**
  - What it is, the two kinds of breakdown, and the V-I characteristic
  - The equivalent circuit, temperature coefficient, and power derating
- **Part 2 — Zener Voltage Regulation**
  - Regulating against a varying input voltage and a varying load
  - From no load to full load, with worked examples
- **Part 3 — Wrap-Up**
  - When to use for a zener, key takeaways, and a look ahead to special-purpose diodes

Today's Learning Objectives

- **By the End of This Lecture, You Will Be Able To**
  - Recognize a zener diode and explain why it operates in reverse breakdown
  - Distinguish zener breakdown from avalanche breakdown
  - Read the zener V-I characteristic and identify V_Z, I_{ZK}, and I_{ZM}
  - Use the ideal and practical (Z_Z) equivalent circuits
  - Apply the temperature coefficient and power derating to find V_Z and $P_D(max)$
  - Analyze zener regulation for a varying input voltage and a varying load

Part 1 — The Zener Diode

Where We Left Off — Why a Zener Diode?

- **V-I Characteristic of a Diode**

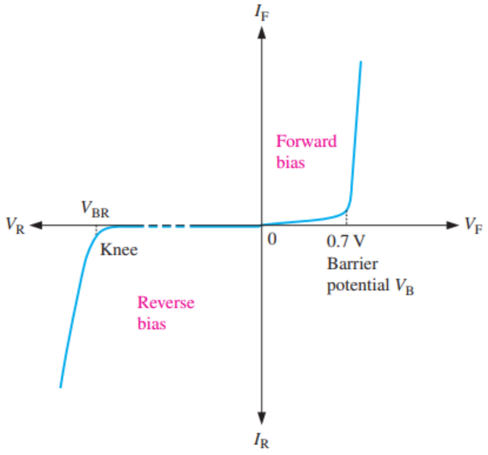


Fig 2.12 — The complete V-I characteristic curve for a diode

- **The Zener Diode Idea**
  - An ordinary diode avoids breakdown; the zener is built to operate in that region on purpose
  - That flat breakdown voltage becomes a stable reference we can regulate with
  - Main uses: voltage references and regulators in power supplies, voltmeters, and instruments

What Is a Zener Diode?

- **A Diode Built for Breakdown**
  - A Zener diode is a silicon pn junction device that is designed to operate in the reverse-breakdown region
  - Its breakdown voltage (V_Z) is set precisely by controlling the doping during manufacture
- **The Symbol**
  - Like a diode, but with a bent cathode line shaped like a 'Z' (for zener)
  - In breakdown, the voltage across it stays almost constant even as the current changes
- **Important Note**
  - In reverse breakdown, the voltage across it stays almost constant even as the current changes drastically – this is key to the Zener diode operation

Fig 3.1 — Zener diode schematic symbol (cathode K, anode A).

The Zener V-I Characteristic

- **Operating Region of Zener Diode**
  - In Forward Bias, it behaves like an ordinary diode
  - The action is in the reverse-breakdown region (the shaded operating area)
- **The Key Feature**
  - Once in breakdown, V_Z is nearly vertical: large current swings produce only a tiny voltage change
  - That near-constant voltage is what makes it a regulator

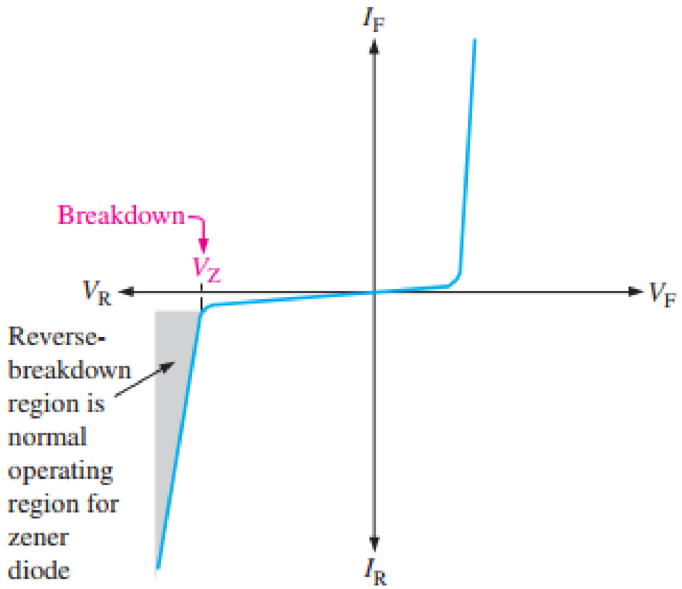


Fig 3.2 — General zener V-I characteristic; the shaded reverse-breakdown region is the operating region.

Two Types of Reverse Breakdown

- **Zener Breakdown (Low Reverse Voltage)**
  - Heavy doping makes a very thin depletion region with an intense electric field
  - The intense electric field pulls electrons loose directly — dominant below about 5 V
- **Avalanche Breakdown (High Reverse Voltage)**
  - The collision-multiplication effect which occurs at high reverse voltages — dominant above about 5 V

- **Operating Voltages**
  - Zener Diodes are commercially available with breakdown voltages from less than 1 V to more than 250 V with specified tolerances from 1% to 20%.

Breakdown Characteristics — I_{ZK}, I_{ZM}, V_Z

- **Reading the Reverse Curve**
  - Reverse current is extremely small until the knee; in breakdown it's called the zener current, I_Z
  - The internal Zener resistance is also called Zener impedance (Z_Z)
  - At breakdown, the Zener impedance decreases as the reverse current increases rapidly
  - V_Z stays essentially constant from the knee up (rising only slightly as I_Z increases)
- **The Two Limits That Bracket Regulation**
  - I_{ZK} — the knee (minimum) current; below it, the voltage collapses and regulation is lost
  - I_{ZM} — the maximum current; above it, power dissipation can destroy the diode
  - So the zener regulates for any I_Z between I_{ZK} and I_{ZM}; V_Z is quoted at a test current I_Z

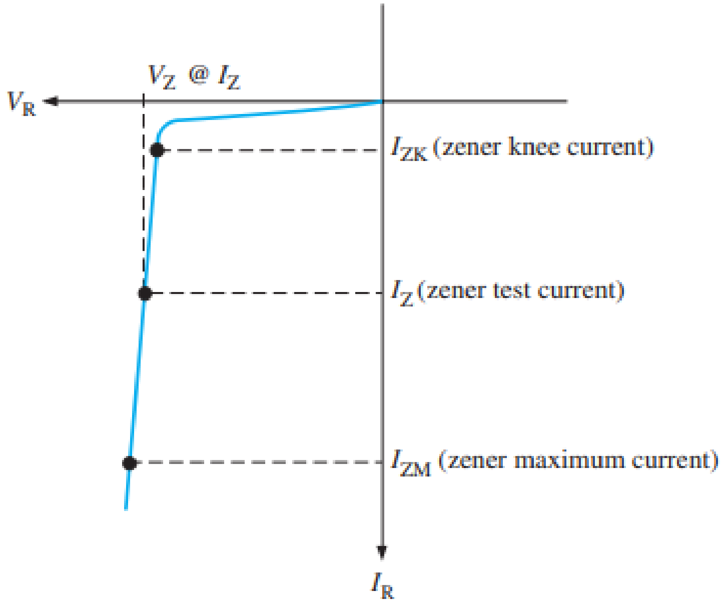


Fig 3.3 — Reverse characteristic: I_{ZK} at the knee, test current I_Z, and maximum I_{ZM}.

Zener Equivalent Circuits

- **Ideal Model (First Approximation)**
  - In reverse breakdown the zener holds an almost constant voltage equal to Vz
  - The ideal model replaces the zener with a constant DC voltage source (a battery symbol — though it generates nothing)
  - The characteristic curve is treated as a vertical line at Vz

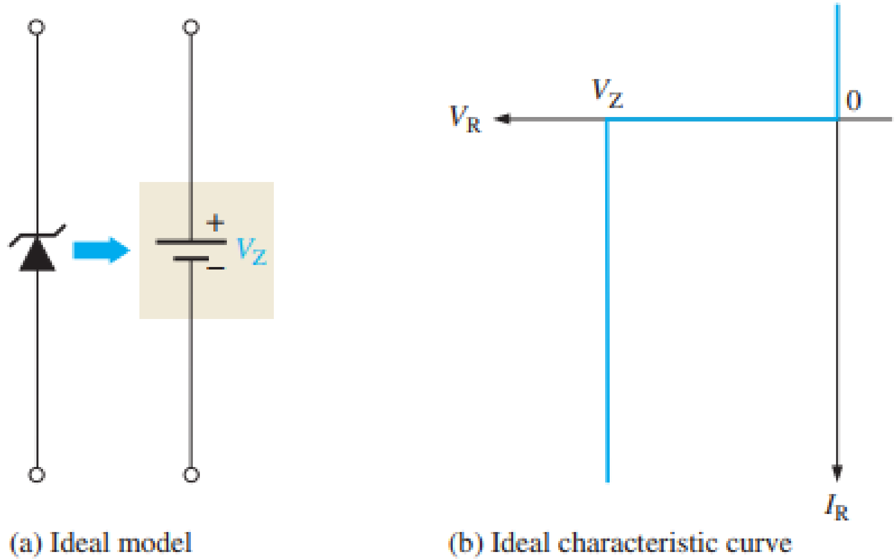


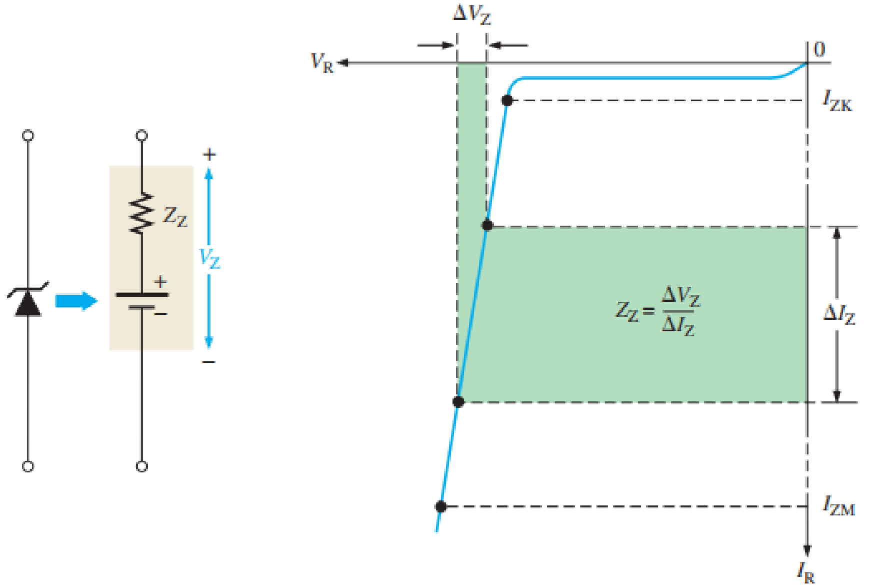
Fig 3.4 — Ideal zener model: a constant VZ and a vertical characteristic curve.

The Practical Zener Model (Second Approximation)

- **The Curve Isn't Perfectly Vertical**
  - A small current change ΔIz produces a small voltage change ΔVz
  - Zener impedance (by Ohm's law):

$$Z_Z = \frac{\Delta V_Z}{\Delta I_Z}$$

  - Zz is specified at the test current; treat it as a small, roughly constant resistance
  - Avoid operating near the knee, where the impedance changes dramatically in that area



(a) Practical model
(b) Characteristic curve. The slope is exaggerated for illustration.

Fig 3.5 — Practical model: the impedance Zz gives the characteristic a slight slope.

Worked Example — Zener Impedance Z_Z

- **Example 3–1**
  - A zener diode exhibits a certain change in V_Z for a certain change in I_Z on a portion of the linear characteristic curve between I_{ZK} and I_{ZM} as illustrated in Figure 3–6. What is the zener impedance?
- **Solution**

$$Z_Z = \frac{\Delta V_Z}{\Delta I_Z} = \frac{50\ \text{mV}}{5\ \text{mA}} = 10\ \Omega$$

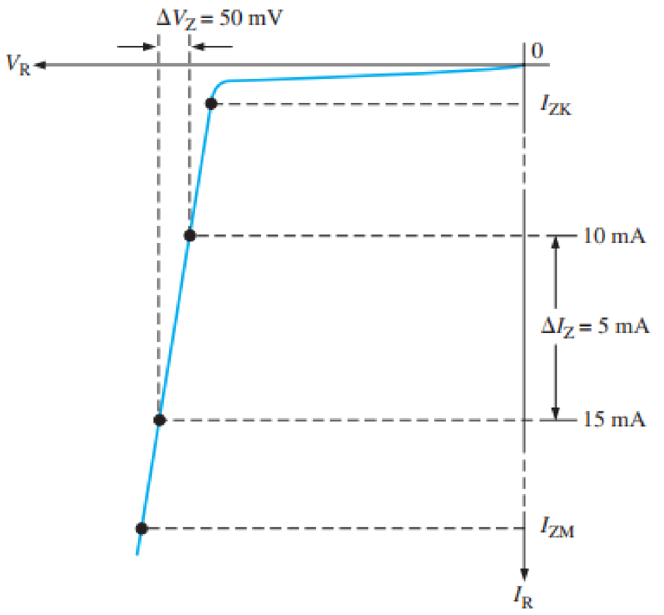


Fig 3.6 — A change ΔVZ over ΔIZ on the linear region gives ZZ.

Temperature Coefficient

- **V_Z Drifts with Temperature**
  - The formula for calculating the change in Zener voltage for a given junction temperature change, for a specified temperature coefficient is

$$\Delta V_Z = V_Z \times TC \times \Delta T$$

  - VZ is the nominal Zener voltage at the reference temperature of 25°C
  - TC is the temperature coefficient which coefficient specifies the percent change in Zener voltage for each degree Celsius change in temperature
  - ΔT is the change in temperature from the reference temperature
- **Sign of TC**
  - Positive TC: V_Z rises as temperature rises
  - Negative TC: V_Z falls as temperature rises
  - In some cases, the temperature coefficient is expressed in mV/°C rather than in %/°C. For these cases, the change in Zener voltage is

$$\Delta V_Z = TC \times \Delta T$$

Worked Example — Temperature Coefficient

An 8.2 V zener diode (8.2 V at 25°C) has a positive temperature coefficient of 0.05%/°C. What is the zener voltage at 60°C?

Solution The change in zener voltage is

$$\Delta V_Z = V_Z \times TC \times \Delta T = (8.2\text{ V})(0.05\%/^\circ\text{C})(60^\circ\text{C} - 25^\circ\text{C})$$
$$= (8.2\text{ V})(0.0005/^\circ\text{C})(35^\circ\text{C}) = 144\text{ mV}$$

Notice that 0.05%/°C was converted to 0.0005/°C. The zener voltage at 60°C is

$$V_Z + \Delta V_Z = 8.2\text{ V} + 144\text{ mV} = \mathbf{8.34\text{ V}}$$

Zener Power Dissipation and Derating

- **How Much Power can a Zener Diode Handle**
  - Zener Diodes are specified to operate at a maximum power called the maximum DC power dissipation

$$P_D = V_Z I_Z$$

  - Datasheets give a maximum, $P_D(max)$ (e.g. 500 mW for a 1N746 Zener Diode, up to tens of watts for big ones)
- **Power Derating**
  - The maximum power dissipation of a Zener diode is specified for temperatures at or below a certain value (e.g., 50 °C)
  - Above a specified temperature, $P_D(max)$ drops by a derating factor (in mW/°C)
  - The maximum derated power can be determined by

$$P_{D(derated)} = P_{D(max)} - (\mathrm{mW/°C})\Delta T$$

Worked Example — Power Derating

- **The Question (Example 3-3)**
  - A certain zener diode has a maximum power rating of 400 mW at 50°C and a derating factor of 3.2 mW/°C. Determine the maximum power the zener can dissipate at a temperature of 90°C.
- **The Solution**
  - Temperature above rating: $\Delta T = 90 - 50 = 40°C$
  - Reduction = 3.2 mW/°C × 40°C = 128 mW
  - Derated power = 400 mW − 128 mW = **272 mW**
- **Related Problem**
  - A certain 50W zener diode must be derated with a derating factor of 0.5mW/°C above 75°C. Determine the maximum power it can dissipate at 160°C.

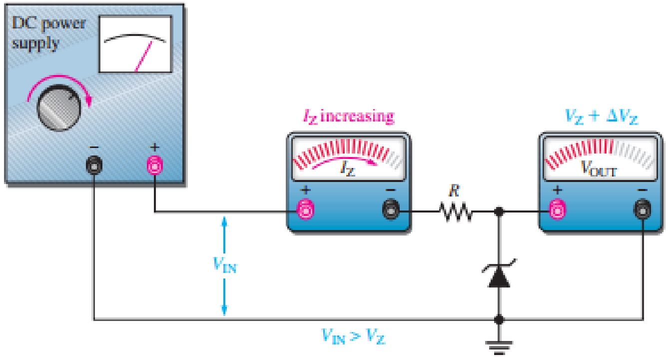
Quick Check #1 — The Zener Diode

- **Question 1**
  - What region is a zener diode designed to operate in, and what sets its breakdown voltage?
- **Question 2**
  - Which breakdown type dominates below ~5 V, and which above?
- **Question 3**
  - What do I_{ZK} and I_{ZM} represent, and why does regulation require staying between them?
- **Question 4**
  - Write the definition of zener impedance Z_Z.
- **Question 5**
  - If a zener has a positive temperature coefficient, what happens to V_Z as it heats up?

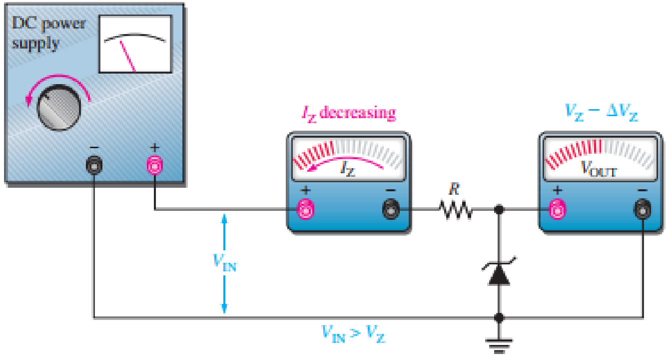
Part 2 — Zener Voltage Regulation

Zener Regulation with a Variable Input Voltage

- **The Circuit**
  - A series current-limiting resistor R feeds the zener; the load sits across the zener
  - As V_{IN} rises and falls (within limits), the zener holds the output near V_Z
- **What Moves**
  - When V_{IN} changes, the zener current I_Z changes proportionally to absorb it
  - The input range is bounded by I_{ZK} (at the low end) and I_{ZM} (at the high end)
  - Zener regulators are simple but inefficient — best for low-current loads



(a) As the input voltage increases, the output voltage remains nearly constant ($I_{ZK} < I_Z < I_{ZM}$).



(b) As the input voltage decreases, the output voltage remains nearly constant ($I_{ZK} < I_Z < I_{ZM}$).

Fig 3.9 — Zener regulation as the input voltage varies.

Worked Example — Input-Voltage Range

- **The Question**
  - A 10 V zener (1N4740A): I_{ZK} = 0.25 mA, $P_D(max)$ = 1 W, with a 220 Ω series resistor. Over what input range does it regulate?
- **The Solution**
  - Max current: $I_{ZM} = P_{D(max)} / V_Z$ = 1 W / 10 V = 100 mA
  - At I_{ZK}: V_R = 0.25 mA × 220 Ω = 0.055 V
    $V_{IN}(min)$ = 10 + 0.055 = 10.055 V
  - At I_{ZM}: V_R = 100 mA × 220 Ω = 22 V

    $V_{IN}(max)$ = 10 + 22 = 32 V
  - So it holds ~10 V out for any input from 10.055 V to 32 V

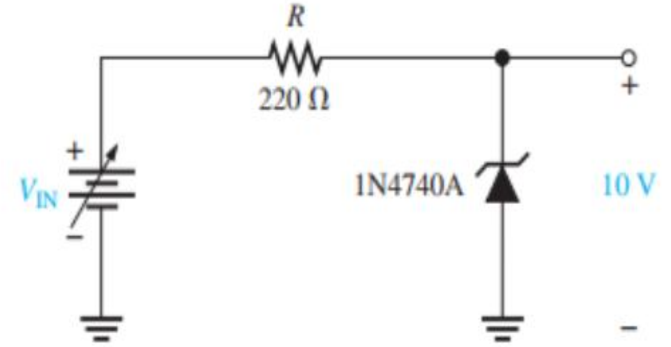


Fig 3.10 — Regulator circuit for the input-range analysis (1N4740A, ideal model).

Worked Example — Input Range Including Z_Z

- **Example 3–5**
  - Determine the minimum and the maximum input voltages that can be regulated by the Zener diode
  - Information from datasheet for a 1N4733A Zener Diode: V_Z = 5.1V at I_Z = 49 mA, I_{ZK} = 1 mA, and Z_Z = 7 Ω at I_Z

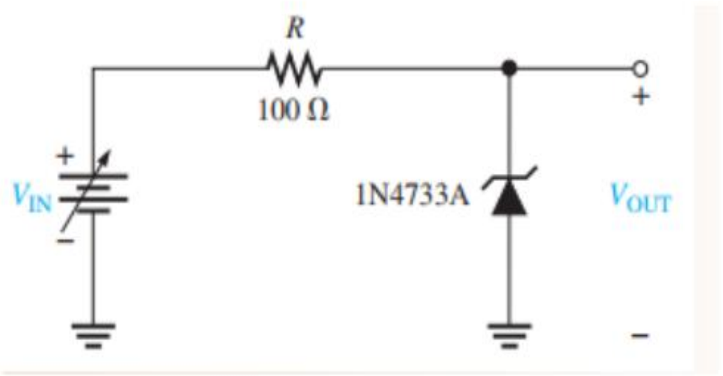


Fig 3.11 — Zener regulator for Example 3–5 (1N4733A).

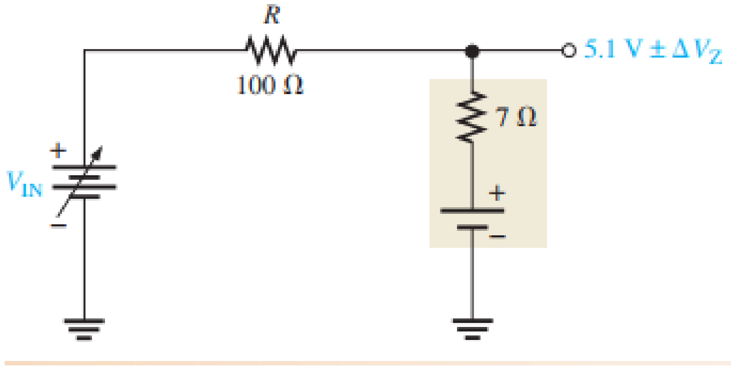


Fig 3.12 — Zener regulator for Example 3–5 Equivalent Circuit

Example 3-5 Solution Continued

At $I_{ZK} = 1$ mA, the output voltage is

$$V_{OUT} \cong 5.1\text{ V} - \Delta V_Z = 5.1\text{ V} - (I_Z - I_{ZK})Z_Z = 5.1\text{ V} - (49\text{ mA} - 1\text{ mA})(7\ \Omega)$$
$$= 5.1\text{ V} - (48\text{ mA})(7\ \Omega) = 5.1\text{ V} - 0.336\text{ V} = 4.76\text{ V}$$

Therefore,

$$V_{IN(min)} = I_{ZK}R + V_{OUT} = (1\text{ mA})(100\ \Omega) + 4.76\text{ V} = \mathbf{4.86\ V}$$

To find the maximum input voltage, first calculate the maximum zener current. Assume the temperature is 50°C or below; so from Figure 3–7, the power dissipation is 1 W.

$$I_{ZM} = \frac{P_D}{V_Z} = \frac{1\text{ W}}{5.1\text{ V}} = 196\text{ mA}$$

At I_{ZM}, the output voltage is

$$V_{OUT} \cong 5.1\text{ V} + \Delta V_Z = 5.1\text{ V} + (I_{ZM} - I_Z)Z_Z$$
$$= 5.1\text{ V} + (147\text{ mA})(7\ \Omega) = 5.1\text{ V} + 1.03\text{ V} = 6.13\text{ V}$$

Therefore,

$$V_{IN(max)} = I_{ZM}R + V_{OUT} = (196\text{ mA})(100\ \Omega) + 6.13\text{ V} = \mathbf{25.7\ V}$$

Zener Regulation with a Variable Load

- **The Circuit**
  - Now the input is fixed, but the load resistor R_L across the zener can change
- **How It Holds**
  - The total current through R_L stays about constant while the zener regulates
  - The zener and the load share that current — whatever the load doesn't take, the zener does
  - Regulation holds as long as I_Z stays between I_{ZK} and I_{ZM}
- **From No Load to Full Load**
  - With the output open ($R_L = \infty$), the load takes no current — all of it flows through the zener
  - Connect R_L: the current splits, part through the load (I_L) and part through the zener (I_Z)
  - As R_L decreases, I_L increases and I_Z decreases (the total through R barely changes)
  - Regulation continues until I_Z falls to I_{ZK} — that's the maximum load (full-load) the zener can support

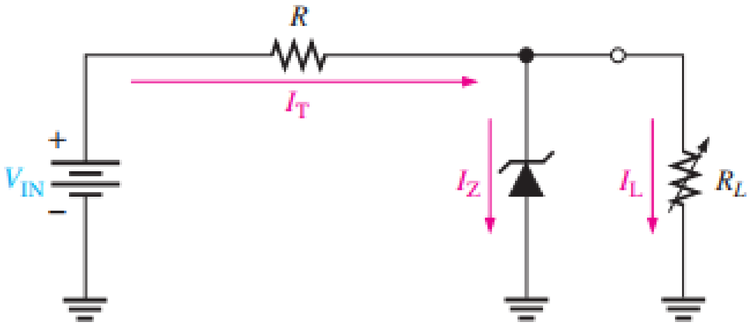


Fig 3.13 — Zener regulation with a variable load.

Worked Example — No Load to Full Load

- **The Question**

Determine the minimum and the maximum load currents for which the zener diode in Figure 3–14 will maintain regulation. What is the minimum value of R_L that can be used? $V_Z = 12$ V, $I_{ZK} = 1$ mA, and $I_{ZM} = 50$ mA. Assume an ideal zener diode where $Z_Z = 0\ \Omega$ and V_Z remains a constant 12 V over the range of current values, for simplicity.

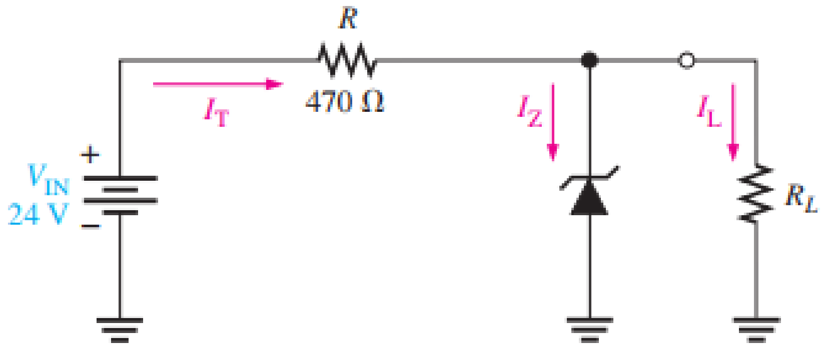


Fig 3.14 — Regulator circuit for the no-load to full-load example.

Worked Example — No Load to Full Load Continued

When $I_L = 0$ A ($R_L = \infty$), I_Z is maximum and equal to the total circuit current I_T.

$$I_{Z(max)} = I_T = \frac{V_{IN} - V_Z}{R} = \frac{24\text{ V} - 12\text{ V}}{470\ \Omega} = 25.5\text{ mA}$$

If R_L is removed from the circuit, the load current is 0 A. Since $I_{Z(max)}$ is less than I_{ZM}, 0 A is an acceptable minimum value for I_L because the zener can handle all of the 25.5 mA.

$$I_{L(min)} = \mathbf{0\ A}$$

The maximum value of I_L occurs when I_Z is minimum ($I_Z = I_{ZK}$), so

$$I_{L(max)} = I_T - I_{ZK} = 25.5\text{ mA} - 1\text{ mA} = \mathbf{24.5\ mA}$$

The minimum value of R_L is

$$R_{L(min)} = \frac{V_Z}{I_{L(max)}} = \frac{12\text{ V}}{24.5\text{ mA}} = \mathbf{490\ \Omega}$$

Worked Example — Load Regulation with Zener Impedance

For the circuit in Figure 3–15:

(a) Determine V_{OUT} at I_{ZK} and at I_{ZM}.

(b) Calculate the value of R that should be used.

(c) Determine the minimum value of R_L that can be used.

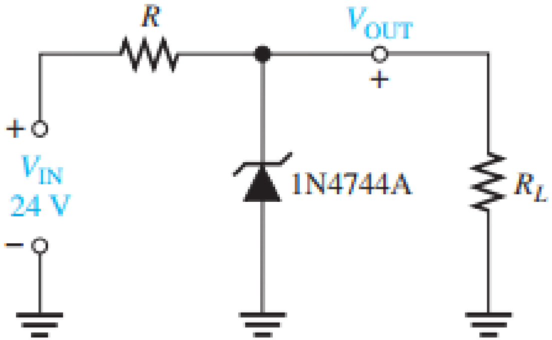


Fig 3.15 — Regulator circuit for the load-regulation example (zener impedance included).

Worked Example Solution — Load Regulation with Zener Impedance

The 1N4744A zener used in the regulator circuit of Figure 3–15 is a 15 V diode. The datasheet in Figure 3–7 gives the following information:
$V_Z = 15\text{ V}$ @ $I_Z = 17\text{ mA}$, $I_{ZK} = 0.25\text{ mA}$, and $Z_Z = 14\ \Omega$.

(a) For I_{ZK}:

$$V_{OUT} = V_Z - \Delta I_Z Z_Z = 15\text{ V} - \Delta I_Z Z_Z = 15\text{ V} - (I_Z - I_{ZK})Z_Z$$
$$= 15\text{ V} - (16.75\text{ mA})(14\ \Omega) = 15\text{ V} - 0.235\text{ V} = \mathbf{14.76\ V}$$

Calculate the zener maximum current. The maximum power dissipation is 1 W.

$$I_{ZM} = \frac{P_D}{V_Z} = \frac{1\text{ W}}{15\text{ V}} = 66.7\text{ mA}$$

For I_{ZM}:

$$V_{OUT} = V_Z + \Delta I_Z Z_Z = 15\text{ V} + \Delta I_Z Z_Z$$
$$= 15\text{ V} + (I_{ZM} - I_Z)Z_Z = 15\text{ V} + (49.7\text{ mA})(14\ \Omega) = \mathbf{15.7\ V}$$

(b) Calculate the value of R for the maximum zener current that occurs when there is no load as shown in Figure 3–16(a).

$$R = \frac{V_{IN} - V_{OUT}}{I_{ZK}} = \frac{24\text{ V} - 15.7\text{ V}}{66.7\text{ mA}} = 124\ \Omega$$

R = **130 Ω** (nearest larger standard value), which reduces I_{ZM} to 63.8 mA.

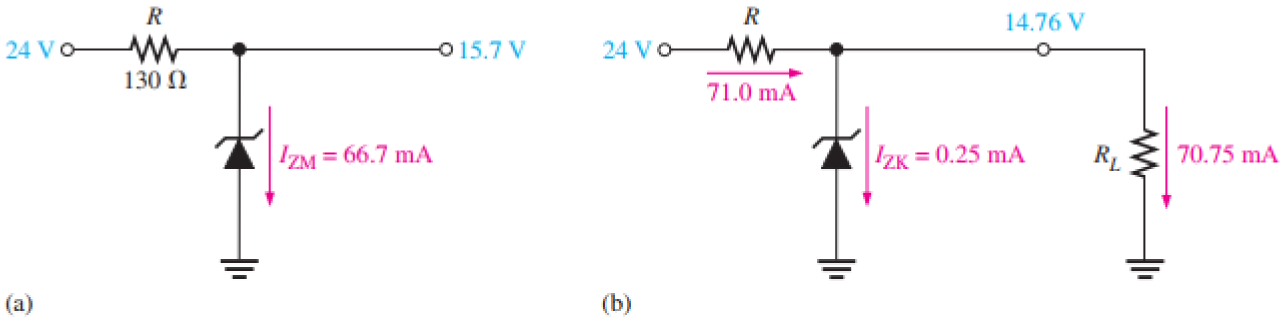
Worked Example Solution — Load Regulation with Zener Impedance

(c) For the minimum load resistance (maximum load current), the zener current is minimum ($I_{ZK} = 0.25$ mA) as shown in Figure 3–16(b).

$$I_T = \frac{V_{IN} - V_{OUT}}{R} = \frac{24\text{ V} - 14.76\text{ V}}{130\ \Omega} = 71.0\text{ mA}$$

$$I_L = I_T - I_{ZK} = 71.0\text{ mA} - 0.25\text{ mA} = 70.75\text{ mA}$$

$$R_{L(\min)} = \frac{V_{OUT}}{I_L} = \frac{14.76\text{ V}}{70.75\text{ mA}} = \mathbf{209\ \Omega}$$



Zener Diode Limits

- **Where the Zener Falls Short**
  - It wastes current and handles only modest loads — its load range is limited by I_{ZK} and I_{ZM}
  - For tighter regulation and larger loads, a transistor or IC regulator is used (built around a zener reference)
  - Zeners also serve as simple limiters/clippers, capping a waveform near V_Z

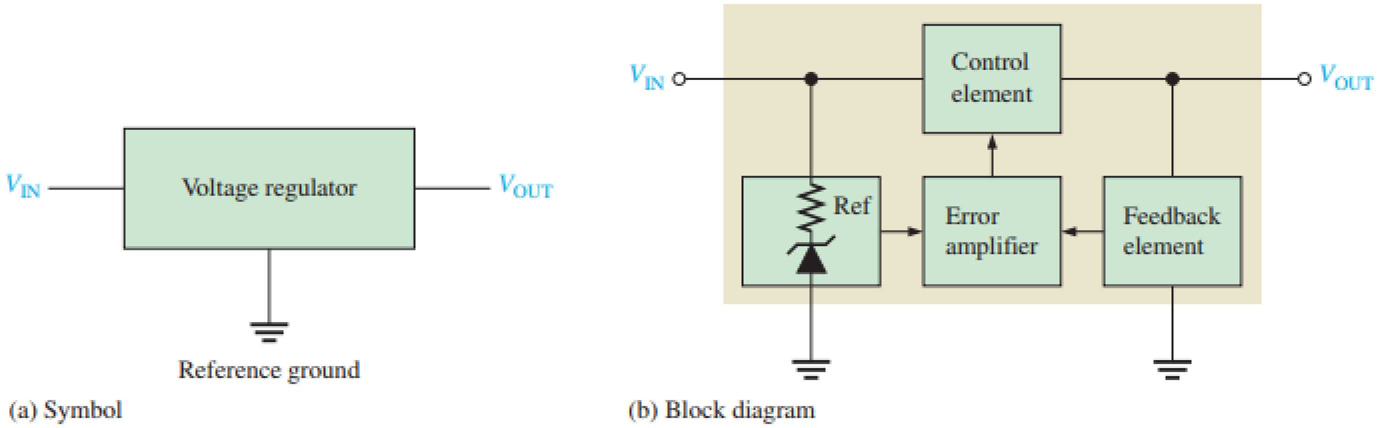


Fig 3.17 — Three-terminal voltage regulators

Quick Check #2 — Zener Regulation

- **Question 1**
  - In input regulation, what changes as V_{IN} varies, and what sets the input limits?
- **Question 2**
  - Why are zener regulators best suited to low-current loads?
- **Question 3**
  - In load regulation, what happens to I_Z as the load current I_L increases?
- **Question 4**
  - What defines the full-load condition for a zener regulator?
- **Question 5**
  - For V_{IN} = 20 V, V_Z = 10 V, R = 220 Ω, what is the total current through R?

Part 3 — Wrap-Up

When to use a Zener Diode

- **Good Fit**
  - You need a stable, low-cost voltage reference (e.g. for a meter or a bias point)
  - You need to regulate a small load where efficiency isn't critical
- **Mind the Limits**
  - Keep I_Z between I_{ZK} and I_{ZM}, and keep P_D below the (derated) maximum
  - Pick V_Z for the voltage you want, and the series R to set the current
- **When to Move On**
  - For higher current or tighter regulation, use an IC/transistor regulator — often with a Zener diode inside as the reference

Key Takeaways from Today

- A zener diode is built to operate in reverse breakdown, where V_Z stays nearly constant as I_Z changes
- Zener breakdown dominates below ~5 V (heavy doping, thin junction); avalanche dominates above ~5 V
- It regulates only between I_{ZK} (knee) and I_{ZM} (max); V_Z is quoted at a test current
- Practical model adds zener impedance: $Z_Z = \Delta V_Z / \Delta I_Z$ — keep away from the knee
- Temperature: $\Delta V_Z = V_Z \cdot TC \cdot \Delta T$. Power: $P_D = V_Z \cdot I_Z$, derated above the rated temperature
- Input regulation: as V_{IN} varies, I_Z absorbs the change (range set by I_{ZK} and I_{ZM})
- Load regulation: Zener and load share the total current; as I_L rises, I_Z falls — full load is reached at I_{ZK}

Coming Up — Week 7: Special-Purpose Diodes

- **Next Topic — Special-Purpose Diodes (Lecture Notes pp. 90–116)**
  - The varactor (voltage-variable capacitance) and its tuning applications
  - Optical diodes: the LED and the photodiode
  - A survey of other special diodes (Schottky, PIN, current-regulator, step-recovery, and more)
- **To Prepare**
  - Keep the diode V-I picture in mind — each special diode tweaks one part of it
- **Reminder**
  - Read pp. 90–116 before class

Thank You — Questions?